H110 — canonical (p^2)-lift defect cocycle

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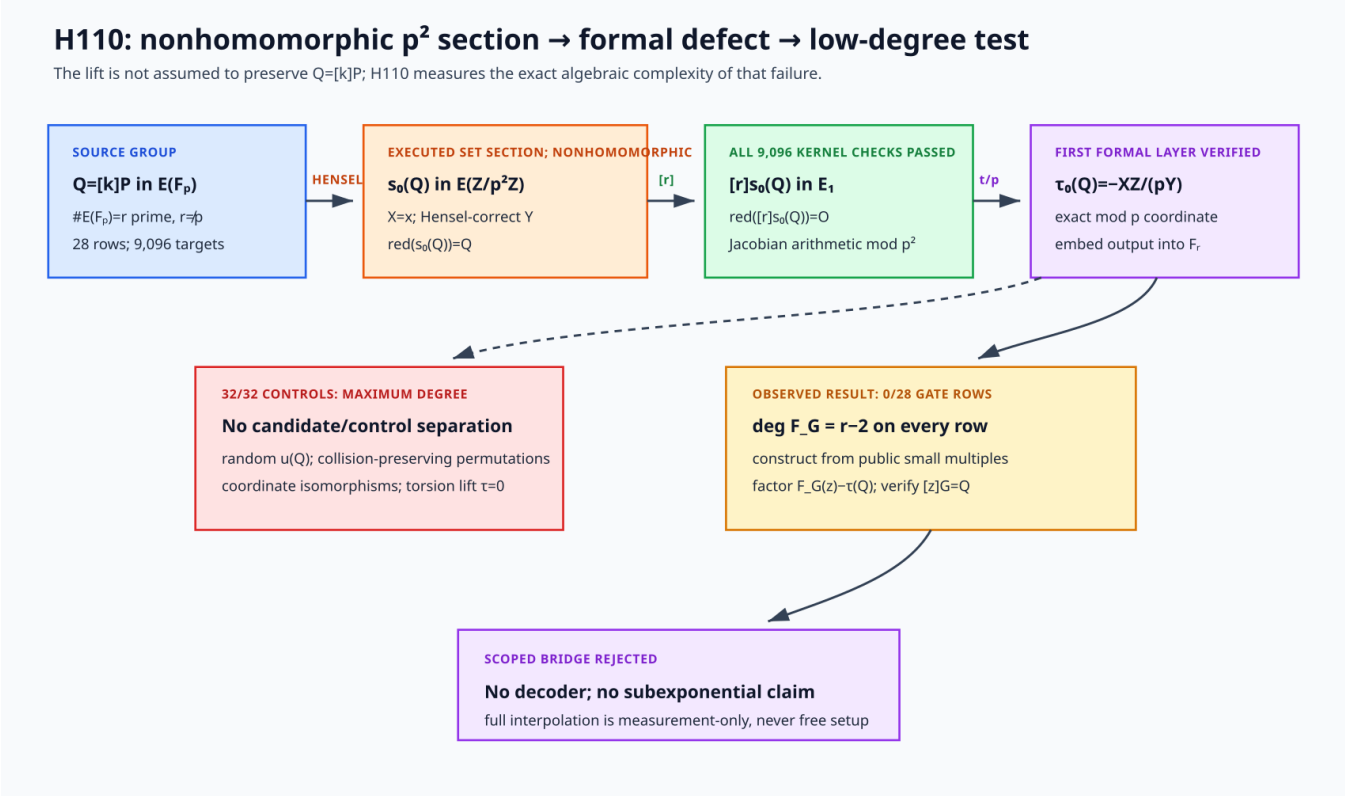


Figure 1: H110 typed transfer and evidence flow

Editable vector source: `h110-p2-lift-defect-flow.svg`.

Requirement-to-evidence contract

- **Mechanism-new relative to H88–H109:** measure the formal-group defect of a deterministic Hensel section into `E(Z/p^2 Z)`; do not add another character, theta word, cover, extension-field phase, or division-polynomial feature family.
- **Transfer audit:** the Hensel section is a set-theoretic section, not a group homomorphism. The experiment must separately check lift existence, specialization, subgroup order, kernel membership, and the formal parameter.
- **Measurement:** use all seven retained prime-order curves, generator relabelings `1,2,3,5`, and every nonzero target. Charge lift construction, projective group operations, interpolation, root search, verification, memory, and advice.
- **Controls:** randomized Hensel sections, collision-preserving output permutations, coordinate-isomorphic models, and the unique torsion-lift correction, plus generic square-root references.
- **Proof:** Lean must prove the abstract section-defect transport identity, certificate-to-candidate implication, EC-verified recovery, and the conditional total-cost composition actually used by any claim.
- **PR:** the canonical and paired reference PRs must be open in draft before the frozen cohort is executed.

Typed correspondence

Let

```
E/F_p : y^2 = x^3 + a*x + b,
#E(F_p) = r,    r odd prime,    r != p.
```

The source is the prime-order group E(F_p)=<P>. The destination is not a replacement group: it is the first formal-neighborhood layer in E(Z/p^2 Z). Reduction gives the exact-sequence fragment

```
E_1(Z/p^2 Z) --> E(Z/p^2 Z) --red--> E(F_p).
```

For a finite point Q=(x,y), use least nonnegative integer representatives and define the canonical coordinate section s_0 by fixing X=x. Write

```
c_Q = (x^3 + a*x + b - y^2) / p  (mod p),
v_0 = c_Q / (2*y)                (mod p),
s_0(Q) = (x, y + p*v_0)          (mod p^2).
```

More generally, a section parameter u in F_p uses

```
X = x + p*u,
v_u = (c_Q + (3*x^2+a)*u) / (2*y)  (mod p),
Y = y + p*v_u.
```

This is a Hensel lift of Q, not a homomorphic lift. Since [r]Q=O, the point [r]s_u(Q) lies in the reduction kernel. In Jacobian coordinates (X_r:Y_r:Z_r) define

```
t([r]s_u(Q)) = -X_r*Z_r/Y_r       (mod p^2),
tau_u(Q) = t([r]s_u(Q))/p          (mod p).
```

The formal logarithm is t + O(t^2), and t^2=0 mod p^2 on this first layer. Thus tau_u is the exact first-layer formal logarithm coordinate.

The correspondence is therefore

```
Q in E(F_p) --set section s_u--> E(Z/p^2 Z)
  --[r]--> E_1(Z/p^2 Z) --t/p--> F_p.
```

Only reduction and multiplication are group maps. The section and the composite tau_u are not assumed additive.

Frozen algorithmic hypothesis

For a generator G, form the exact word

```
w_G(k) = integer representative of tau_0([k]G),  1 <= k < r,
```

and embed its values into F_r. Let F_G(z) be the unique polynomial of degree < r-1 over F_r satisfying F_G(k)=w_G(k).

The H110 prediction is that the canonical section defect has degree

```
deg(F_G) <= D(r) = ceil(log2(r))^2
```

on every frozen row, while matched controls do not. If so, the first D(r)+1 public small multiples construct F_G; given a target Q, compute tau_0(Q), factor F_G(z)-tau_0(Q) over F_r, and retain the unique root whose scalar multiple equals Q. At most D(r) EC validations are charged.

For the bounded experiment, enumerate all roots in F_r; this is a toy verification route, not the claimed asymptotic recovery routine. A positive algorithmic result additionally requires a proved polynomial-factorization bound and a uniform theorem establishing the degree bound.

Frozen cohort and controls

1. Run curves E101, E127, E149, E211, E283, E503, and E907, generator multipliers 1,2,3,5, and every 1 <= k < r.
2. Check the lifted curve equation modulo p^2, reduction to [k]G, the projective curve equation for [r]s_u(Q), reduction to identity, divisibility of the formal parameter by p, and agreement with a separately composed affine/formal calculation wherever affine denominators are units.
3. Compute the complete interpolation degree, the frozen prefix certificate, collision multiplicities, root counts, operation counts, and all ties.

4. On each of primary E101 and E127, run eight deterministic random-section controls `u_seed(Q)` and eight collision-preserving random permutations of the canonical output word. These 32 controls receive the identical degree calculation and gate.
5. On the primary generators, repeat the canonical word after the coordinate isomorphism

   `(x,y,a,b) -> (lambda^2*x, lambda^3*y, lambda^4*a, lambda^6*b)`

   with the least `lambda>1` in `F_p`; report whether the exact degree is invariant.
6. For every target, use `u=0` and `u=1` to solve the first-layer linear correction that makes `tau_u(Q)=0`, then directly check the resulting torsion lift. This is a structural control: the homomorphic torsion section kills the proposed observable and does not itself solve ECDLP.
7. Execute the root/EC decoder only for rows meeting `deg(F_G)<=D(r)`.

Stopping is exhaustive over the frozen matrix. Any lift, reduction, kernel, formal-parameter, section-linearity, interpolation, root, or EC mismatch invalidates the run.

Cost and promotion gates

Report

```
C_total = C_lift + C_[r] + C_interpolation
        + C_factor_or_root + C_EC_verify + C_checks.
```

Units are modular additions/multiplications/inversions modulo `p^2` and `p`, Jacobian additions/doublings, interpolation field operations over `F_r`, roots, EC validations, stored field elements, and advice coefficients. No fixed-generator table may be omitted from cost.

H110 qualifies as a generic subexponential algorithm only if:

1. every generic row has an exact degree at most `D(r)` under a public rule;
2. the canonical section and defect are computable in `poly(log p,log r)`;
3. polynomial construction and factorization, recovery, verification, setup, memory, and advice satisfy a proved subexponential total bound;
4. the degree theorem is uniform, not learned from the full scalar word;
5. Lean proves the interfaces and cost composition used by the claim.

The frozen full interpolation is a measurement instrument. Its linear or quadratic work cannot be counted as the proposed algorithm.

Prior-art boundary

The bounded screen located:

- Smart, "The Discrete Logarithm Problem on Elliptic Curves of Trace One", for the anomalous `r=p` case;
- Huang, Kueh, and Tan, "Lifting Elliptic Curves and Solving the Elliptic Curve Discrete Logarithm Problem";
- Silverman, "Lifting and Elliptic Curve Discrete Logarithms", especially the failure of arbitrary local nontorsion lifts to preserve `Q=[k]P` and the unique prime-to-`p` torsion lift;
- Yasuda, "The Elliptic Curve Discrete Logarithm Problems over the p-adic Field and Formal Groups".

Those sources make lifting itself established prior art and identify the relation-preservation obstruction. The bounded screen did not locate this specific complete test of the canonical section's first-layer defect as a uniform low-degree scalar polynomial. Status at freeze: mechanism-distinct within this repository; external novelty not established.

Preregistered interpretation

- A degree above `D(r)` rejects only the frozen canonical-section polynomial bridge.
- A low degree on toy rows is not a generic theorem or speedup.
- Random-section-like behavior indicates coordinate-section structure rather than an EC-specific scalar oracle.
- The torsion-lift zero control cannot be promoted into a formal-log attack: its formal observable is identically zero.

Executed experiment

The paired draft gates were opened as isogeny-labs/crypto#3111 and isogeny-labs/cryptanalysis#1346 before implementation. The frozen coefficient section, cohort, controls, and stopping rule were not changed.

Every one of the 9,096 canonical target records passed:

- the Hensel-lift curve equation modulo `p^2` and reduction back to `Q`;
- two independently split computations of `[r]s_0(Q)`;
- the projective curve equation, reduction-kernel check, unit-`Y` check, and divisibility of the formal parameter by `p`;
- nonzero first-layer section slope from `u=0` and `u=1`;
- direct reevaluation of the solved torsion correction, with `tau_u(Q)=0`.

The evaluator executed 36,384 lift constructions, mixed kernel additions, and formal extractions. The canonical rows used 72,768 scalar multiplications, 296,560 affine additions, 521,064 affine doublings, 6,456,536 counted modular additions, 7,820,632 modular multiplications, and 890,392 inversions. The recorded canonical evaluation and interpolation portions took 2.4371 s and 0.3836 s, respectively; these are diagnostic container timings, not isolated benchmarks or scaling evidence.

The algorithmic prediction failed on every row. For `r-1` sampled scalar positions the largest possible interpolation degree is `r-2`, and every one of the 28 candidate rows attained it:

curve	r	observed degree (all four generators)	frozen gate
E101	83	81	49
E127	109	107	49
E149	139	137	64
E211	223	221	64
E283	281	279	81
E503	499	497	81
E907	947	945	100

The candidate values had maximum collision buckets of four or five, so the failure is not merely a constant word. The gate-limited Newton prefixes matched only 50–103 targets per row, close to their available coefficient counts and far from exactness.

All eight random-section and all eight collision-preserving permutation controls on each of E101 and E127 also had maximum degree: 81 and 107, respectively. Coordinate-isomorphic canonical words were byte-distinct from the originals but retained those same maximum degrees. Thus no control separated from the candidate and zero root/EC decoders opened.

A second complete run reproduced scientific-content SHA-256 `3397823fe3f55fe008797bb87b31324b7ae33efb5699fe479e8e58f5`. An independently composed standard-library verifier recomputed all Hensel and group arithmetic, all 28 degrees, all 32 matched controls, and the two coordinate-isomorphic controls. The committed result SHA-256 is `7d6c390d0156a4ed101bddd0969132a85505fa884282ef505dc4621b7852124e`. Lean 4.19 proves the abstract section-defect transport, torsion-correction, verified-candidate, uniqueness, and total-cost interfaces. It does not prove the rejected low-degree hypothesis.

Result: the first formal layer is executable and structurally well checked, but the coordinate-fixed section defect is maximally complex under the frozen scalar-polynomial representation. This closes H110's proposed bridge. It does not rule out other lift-derived invariants or establish a generic ECDLP lower bound.